

Relevance Rules Confound PDF Chunker Evaluation

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Abstract

Retrieval-augmented generation over PDFs depends on how a document is split into chunks, and chunkers are compared on leaderboards. We show that those leaderboards are underdetermined by their own design. Scoring seven PDF chunkers on *identical chunks* over 95 documents and 3,558 human-annotated questions, we find that the ranking changes with the rule used to decide whether a retrieved chunk counts as relevant: one tool ranks 1st of 7 under span and region relevance and 6th of 7 under page relevance, and the tool that displaces it shows the mirror pattern. No rule is neutral to chunk size, so a single-rule comparison entangles a ranking with a size preference. We therefore report three rules, and add an evaluation that uses no rule at all, scoring chunk boundaries against publisher-authored JATS section structure for 134 PubMed Central papers with Pk and WindowDiff. Our instrument is DOCSTRUCT, a local, deterministic, structure-aware chunker that invokes no language model on the parse path.

1 Introduction

A retrieval-augmented generation (RAG) system (Lewis et al., 2020) over PDFs is only as good as the units it retrieves. Those units, chunks, come from a pipeline that in most deployments is structurally blind twice over. A parser converts the PDF into a character stream and discards the two-dimensional layout that told a human reader where a section began and which column to read next. A splitter then cuts that stream every N characters, with a separator preference list as its only concession to structure. Neither stage has a representation of “section”, “table” or “column”.

The consequences appear in any real corpus. `page.extract_text()` welds a two-column page into unreadable lines, a chunk boundary lands mid-sentence because a character budget expired, a table is severed from the header row that gives its

columns meaning. The parser’s own metrics cannot see any of this, because it was asked to preserve characters and never structure.

Recent work responds by making segmentation semantic: a language model judges where topics change (Duarte et al., 2024), or an embedding model clusters sentences into coherent groups (Smith and Troynikov, 2024). These methods improve on fixed-size splitting and carry three costs that are rarely priced. They are non-deterministic, so the same PDF can yield different chunks on consecutive runs. They spend one model call per candidate boundary on every document. And their boundaries are a property of a model checkpoint, so an upgrade silently invalidates an index. In a regulated or auditable pipeline, the setting where document structure matters most, those three costs are disqualifying.

DOCSTRUCT takes the opposite position. Structure in a born-digital PDF is a geometric fact, recoverable from the file’s own primitives, so a chunker can read it directly with no model on the parse path. Section 3 describes the pipeline.¹

Building DOCSTRUCT required us to answer a prior question: how does anyone know a chunker is good? The standard answer is a retrieval leaderboard, so we ran one, and found its result undefined until one more choice is fixed, a choice the literature makes silently. Scoring a retrieved chunk as relevant requires a rule, and different reasonable rules produce different winners from the same chunks.

Contributions.

1. A measured inversion. On 95 documents and

¹Code, every report cited here, the corpus-fetch scripts, the manifests and SHA-256 checksums, and the script that generates every figure from those reports are at <https://github.com/CandyButcher27/DocStruct>. Corpora are fetched rather than redistributed, so each is reconstructible byte for byte; `scripts/verify_determinism.py` reproduces §5.3. The library installs as `pip install docstruct-rag`.

076	3,558 human questions, DOCSTRUCT ranks	125
077	1st of 7 under span and region relevance and	126
078	6th of 7 under page relevance, on chunks that	
079	never changed between the three runs (§5.1).	
080	We could find no prior chunking evaluation	
081	that reports more than one rule. A sweep of	
082	the region threshold (§5.1) shows the effect be-	
083	longs to the rule rather than to a badly chosen	
084	constant.	
085	2. An evaluation of chunkers that uses no rel-	
086	evance rule. We score chunk boundaries	
087	against the section boundaries publishers	
088	wrote in JATS XML for 134 PubMed Central	
089	papers, using the standard segmentation error	
090	rates Pk and WindowDiff (§5.2). No embed-	
091	der, no rule, and gold written for an unrelated	
092	purpose before this benchmark existed.	
093	3. Evidence that a chunker with no model on its	
094	parse path is competitive. DOCSTRUCT is the	
095	instrument for the two results above (§3), and	
096	its determinism is measured on 95 documents	
097	in two independent processes (§5.3), a row no	
098	model-segmented chunker can fill. We claim	
099	the property, not the architecture: §5.4 reports	
100	that our own vision detector fails to earn its	
101	place.	
102	4. Results against our own system, reported in	
103	the main text (§5.4). The vision detector	
104	is null on every corpus we can check, and	
105	geometry-only beats the hybrid on the struc-	
106	tural metric. Our context bill is $3.9\times$ the lean-	
107	est baseline, which leaves that baseline $3.6\times$	
108	ahead of us on rank quality per token. Our	
109	word coverage is 0.9632 against LangChain’s	
110	1.000, at the highest duplication of the seven.	
111	And a widely used financial corpus turns out	
112	to be unusable as a retrieval benchmark under	
113	a fixed-embedder protocol, where a published	
114	paper reports MRR 0.700–0.844 on the same	
115	files.	
116	2 Related Work	
117	Parse fidelity benchmarks. OmniDocBench	
118	(Ouyang et al., 2025) and READoc (Li et al., 2025)	
119	evaluate whether a converter reproduces a docu-	
120	ment’s text, tables and reading order, using nor-	
121	malised edit distance, TEDS and vocabulary F1.	
122	Docling (Auer et al., 2024) sits in the same family.	
123	These measure the parse and stop before chunking,	
124	and therefore before retrieval. Our work is com-	
	plementary. We report no parse-fidelity number,	
	which we list as a limitation.	
	Chunking strategies. LumberChunker (Duarte	
	et al., 2024) uses a language model to locate	
	topic shifts. Chroma’s technical report (Smith and	
	Troynikov, 2024) introduces a cluster-based seman-	
	tic chunker and a token-level evaluation vocabulary	
	(IoU, Precision _Q). Jimeno-Yepes et al. (2024) show	
	that element-type chunking beats fixed-size split-	
	ting on financial reports, the closest prior claim to	
	ours. Śmigielski et al. (2026) ask whether semantic	
	chunking earns its compute and largely conclude	
	that it does not, which supports our determinism ar-	
	gument. Bennani and Moslonka (2026) report that	
	overlap gives little benefit and identify a “context	
	cliff” past roughly 2,500 tokens; Bhat et al. (2025)	
	find that chunk size dominates other choices. A	
	2026 wave of structure-aware chunkers shares our	
	positioning and post-dates our design.	
	End-to-end evaluation and segmentation met-	
	rics. Closest to our setting, El Bachyr et al.	
	(2026) benchmark parser and chunker combina-	
	tions for financial QA, and OHR-Bench (Zhang and	
	Others, 2025) measures how OCR noise propagates	
	into RAG while supplying the human-annotated	
	corpus we adopt. Both report a single relevance	
	rule, and §5.1 shows the choice of rule can reverse	
	a ranking. Pk (Beeferman et al., 1999) and Win-	
	dowDiff (Pevzner and Hearst, 2002) are the stan-	
	dard error rates for linear text segmentation; to our	
	knowledge they have not been used to evaluate	
	RAG chunkers against publisher-authored struc-	
	ture, which is what §5.2 does.	
	3 Method	
	3.1 Problem setting	
	A chunker maps a PDF to an ordered list of chunks.	
	Each chunk carries text, a page number, a position	
	in reading order, and a section path: the stack of	
	headings above it. Downstream, an embedder and	
	a retriever consume that list and nothing else, so	
	every decision the chunker makes is visible to re-	
	trieval and invisible to any metric computed on the	
	parse alone.	
	Our design constraint is that the mapping be	
	a pure function of the file: the same PDF must	
	produce the same chunks in any process, on any	
	day, with no network and no model call on the	
	parse path. That rules out the semantic chunkers of	

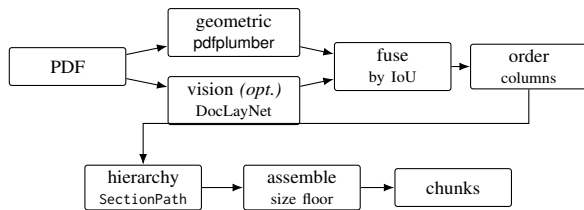


Figure 1: The DOCSTRUCT pipeline. Two detectors read disjoint evidence from the same page and are fused into one block sequence; the remaining stages are deterministic transforms of that sequence. Removing the vision detector yields DOCSTRUCT-GEO, which we report throughout as a separate system.

§2 by construction, and makes the measurement of §5.3 possible at all.

3.2 Pipeline

Figure 1 shows the five stages and the object each one produces. Every threshold lives in a single config.py, and each tuned constant carries a comment naming the measurement that chose it.

Stage 1: two mutually blind detectors. The geometric detector reads PDF primitives through pdfplumber, namely character positions, font names and sizes, and ruling lines. It proposes blocks from font-size clustering, whitespace runs and ruling geometry. It is exact where the PDF is well formed and blind where structure is implied visually, since a borderless table has no rules to find. The vision detector, a YOLOv8m trained on DocLayNet (Pfzmann et al., 2022), proposes the same block types from page rasters and appearance alone. Each detector is blind to what the other reads directly, which is what makes their agreement informative. The vision detector is optional; with no weights present the pipeline runs geometry-only, reported throughout as DOCSTRUCT-GEO.

Stages 2 and 3: fusion and reading order. Proposals are matched by intersection over union and merged into one block sequence, each block carrying a confidence derived from whether one or both detectors saw it. Coordinates are top-left throughout, matching pdfplumber. The scale factor for a unilateral detection and the clamps on the resulting confidence are inherited from a prototype and remain marked # unvalidated, because we have no annotated layout gold at the scale needed to tune them. Blocks are then ordered by a column-aware sweep that infers column bands from the distribution of block x positions and reads each band top

to bottom before moving right; we evaluated recursive XY-cut and rejected it, because it cost retrieval accuracy without simplifying the sweep it replaced.

Stage 4: section hierarchy. Headings are classified by font size and weight relative to body text, and each block receives a SectionPath, the stack of headings above it, which travels with the chunk into the index.

Stage 5: chunk assembly. Text is extracted per block with a font-scaled word-gap tolerance, which recovers spacing that a fixed tolerance mis-splits on display type. Chunks are then assembled by walking blocks in reading order and closing a chunk when a structural boundary is reached and the accumulated content exceeds MIN_CHUNK_TOKENS. Tables are emitted as their own chunks with their header rows attached.

That size floor is the most consequential parameter in the system. Flushing on every structural boundary alone produces chunks too small to carry their own evidence, so the floor trades boundary fidelity for chunk self-sufficiency. It is also the parameter that the relevance-rule result of §5.1 turns out to be measuring, once that section separates a ranking from a size preference.

4 Evaluation Setup

4.1 Protocol

Every tool in every table below runs through one harness, in which the chunker is the only variable. Held fixed across all seven: the embedder (all-MiniLM-L6-v2, Reimers and Gurevych, 2019), the lexical retriever (BM25, Robertson and Zaragoza, 2009), their combination by reciprocal rank fusion at $k=60$ (Cormack et al., 2009), and a retrieval depth of $k=5$. We report MRR, nDCG@5 (Järvelin and Kekäläinen, 2002), Recall@5 and Precision@1, each with a bootstrap 95% confidence interval, and paired bootstrap p -values against DOCSTRUCT (Efron and Tibshirani, 1993). Pairing matters here, because comparing overlapping marginal intervals is the standard way to wrongly call a real difference insignificant.

4.2 Corpora

Table 1 lists the three corpora and the role each one plays. All gold in this paper is human-authored or publisher-authored. We report no corpus of our own: a model-generated benchmark over a scrape

Table 1: The three corpora. OHR-Bench carries the retrieval leaderboard, PMC carries a rule-free structural evaluation, and FinanceBench is reported as a negative result (§5.4) rather than as a leaderboard.

Corpus	Docs	Items	Gold
OHR-Bench	95	3,558 q	human annotators
PMC	134	≈ 25 sec./doc	publisher JATS
FinanceBench	84	189 q	financial analysts

we control would be the weakest evidence we could offer.

OHR-Bench (Zhang and Others, 2025) is our primary external benchmark: 95 born-digital documents across law, manuals, finance and academia, with 3,558 human-authored questions and human-annotated evidence spans. It carries the headline result because its gold is human, external, and pre-dates our system.

PubMed Central supplies 134 open-access papers from seven journals, each fetched with its publisher’s JATS XML through the Europe PMC RESTful web service (Europe PMC Consortium, 2026). The XML carries the real section hierarchy, authored by the publisher and never touched by a chunker. This is the strongest tool-agnostic gold available to us, because it was written for an unrelated purpose before this benchmark existed.

FinanceBench (Islam et al., 2023) was intended as a second external leaderboard. §5.4 reports why it cannot serve as one under our protocol.

4.3 Relevance rules

Deciding whether a retrieved chunk is relevant requires a rule, and the choice is not innocuous. We implement three. Under **span**, a chunk is relevant when it contains the gold answer span verbatim, ignoring whitespace. Under **region**, when it overlaps the gold evidence region by at least $\text{RELEVANCE_REGION_MIN_OVERLAP} = 0.7$. Under **page**, when it comes from the page the gold is annotated on.

The size preferences follow from the definitions. A longer chunk is more likely to contain any given span, so span rewards large chunks. A shorter chunk is more likely to sit entirely on one page, so page rewards small ones. Region is designed to be size-tolerant, and we measure that rather than assuming it: across the seven tools, Spearman’s ρ between mean chunk width and MRR is +0.57 under span and -0.68 under page, and +0.04 under region. With seven tools none of the three is

Table 2: OHR-Bench: 95 documents, 3,558 human questions, seven chunkers, MRR under three relevance rules on *identical chunks*. Rank in parentheses; ctx is context words returned per query.

Tool	page	span	region	ctx
DOCSTRUCT	0.600 (6)	0.706 (1)	0.666 (1)	2194
DOCSTRUCT-GEO	0.470 (7)	0.705 (2)	0.657 (2)	2328
pymupdf4llm	0.668 (4)	0.699 (3)	0.604 (3)	2424
unstructured	0.795 (1)	0.654 (5)	0.601 (5)	560
langchain	0.756 (2)	0.641 (7)	0.603 (4)	638
llamaindex	0.729 (3)	0.648 (6)	0.589 (6)	1430
llamaindex_sem	0.652 (5)	0.654 (4)	0.575 (7)	4698

significant, so we read the signs and not the magnitudes; the sign reversal between span and page is the effect this paper is about, and region sitting at zero is the rule behaving as designed.

Reachability is a ceiling, not a score. Before running any leaderboard we measure what fraction of a corpus’s gold is findable in the PDF’s own text at all. The number is identical for every tool, so it says whether a rule can measure the corpus rather than which tool is better. On OHR-Bench, 80.2% of gold is span-reachable. The measurement carries a circularity warning: once gold occupies a large share of its page, as it does on FinanceBench at 69%, span and region reachability approach 100% by construction and prove nothing.

5 Results

We ask four questions in turn. Does the relevance rule change the ranking (§5.1)? Does that finding survive the region threshold (§5.1)? Does a chunker ranking exist at all without a relevance rule (§5.2)? And is the determinism claim true when tested (§5.3)? Section 5.4 then reports where DOCSTRUCT loses.

5.1 The relevance rule changes the ranking

Table 2 scores the seven chunkers under all three rules. The three runs share identical chunks: chunk counts and mean chunk widths are equal across the modes, because chunking came from a warm cache and only scoring changed. The relevance rule is therefore the only variable in the table.

DOCSTRUCT ranks 1st of 7 under span and region and 6th of 7 under page. Unstructured is the mirror image, 1st under page and 5th under both others. The gaps are not noise. Under region, DOCSTRUCT beats all five external tools significantly; under span it beats four of five, with pymupdf4llm

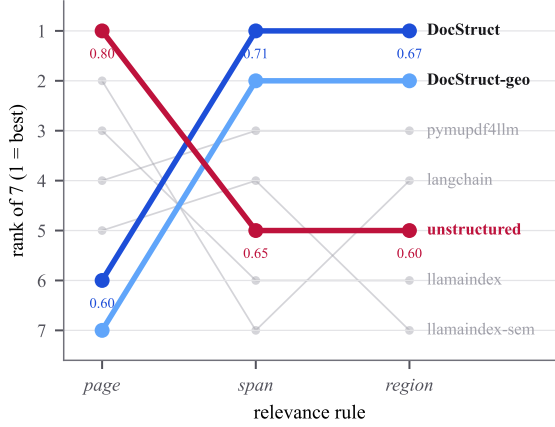


Figure 2: Rank under each relevance rule, same chunks throughout. DOCSTRUCT runs 6th, 1st, 1st and unstructured runs 1st, 5th, 5th. Only the rule that decides whether a retrieved chunk counts as relevant changes between the columns.

inside the noise ($p = 0.23$); under page it loses to four of five.

The mechanism is chunk size. Unstructured emits the smallest chunks in the field, at 560 context words per query, and wins the rule that rewards small chunks. DOCSTRUCT emits large structural units, at 2,194 words, and wins the two rules that reward them. The clearest case is internal to our own system: under page relevance DOCSTRUCT-GEO scores 0.470 against the hybrid’s 0.600, and the entire gap comes from geometry-only emitting 5,810 chunks against the hybrid’s 9,080. Read as a component ablation that would be a +0.13 gain from the vision detector; it is an artefact of chunk count.

The consequence for the field is that a chunking leaderboard reporting one relevance rule has reported a ranking and a size preference entangled together. We publish all three rules, and recommend that the shape of a corpus’s gold choose the rule: span where gold marks a verbatim sentence, region where it marks a block, and page where the annotation carries no finer locator than a page number, which is the case for much human-annotated document QA and is the reason we report page relevance rather than discarding the rule we lose under.

One scope note. Every number here comes from a single retrieval configuration (§4.1), so what we establish is that the rule changes the ranking under that protocol. Whether the inversion survives a stronger embedder, dense-only retrieval or a differ-

Table 3: OHR-Bench MRR under *span* relevance by domain. Our aggregate win is carried by law and manuals; we place behind an external tool on both academic and finance. Question counts in the header.

Tool	acad. (631)	fin. (736)	law (1515)	manual (676)
DOCSTRUCT	0.453	0.514	0.866	0.793
DOCSTRUCT-GEO	0.457	0.501	0.866	0.797
pymupdf4llm	0.511	0.468	0.857	0.773
unstructured	0.515	0.447	0.794	0.706
llamaindex_sem	0.286	0.535	0.841	0.708
llamaindex	0.202	0.520	0.837	0.782
langchain	0.193	0.493	0.831	0.792

ent retrieval depth is open, and is the experiment we would run next.

The obvious attack on the region column is that RELEVANCE_REGION_MIN_OVERLAP = 0.7 was an unvalidated constant. We dumped the continuous overlap behind every retrieved chunk in one run and re-scored offline at every threshold, so the chunks are identical at every point on the curve; the dumping run reproduces the published leaderboard to a maximum MRR drift of 0.0002. A DOCSTRUCT variant places 1st at every threshold, with a margin of +0.045 to +0.062 over the best external tool across 0.4 to 1.0, while the field below us reorders constantly. Two caveats: 0.7 is where our margin happens to peak, at +0.0619, which we state rather than leave to a reader; and the low end is uninformative, since at threshold 0 every chunk is relevant by definition. The ten points are re-scorings of one run, so they are ten views of the same data and not ten independent confirmations. What they establish is insensitivity to the constant.

Table 3 splits that span column by domain, and the aggregate hides two losses: we trail unstructured on academic documents and llamaindex_sem on finance, and the win is carried by law and manuals. Splitting by evidence source is bleaker for everyone: where gold sits in a table, the best of the seven scores 0.307 and we score 0.288.

5.2 Section boundaries, with no retriever

Every result so far passes through an embedder and a relevance rule; this one passes through neither. We compare each chunker’s boundaries against the section boundaries the publisher wrote in JATS XML, using the error rates Pk (Beeferman et al., 1999) and WindowDiff (Pevzner and Hearst, 2002).

DOCSTRUCT-GEO leads both metrics, on a cor-

Table 4: Section-boundary agreement against publisher JATS gold, 134 PubMed Central papers. Pk and WindowDiff are error rates (lower is better). Straddle rate is descriptive. Unstructured’s row covers 99 documents, because it hard-failed on 35.

Tool	WinDiff	Pk	Strad.	Chunks
DOCSTRUCT-GEO	0.423	0.342	0.513	26.8
pymupdf4llm	0.480	0.449	0.573	17.7
DOCSTRUCT	0.482	0.353	0.439	37.5
llamaindex_sem	0.534	0.513	0.189	29.1
llamaindex	0.695	0.598	0.366	42.7
langchain	0.879	0.620	0.220	85.6
unstructured*	0.893	0.603	0.182	106.9

* 99 of 134 documents; 35 hard failures.

pus whose gold was written by publishers for their own typesetting. Note which system leads: geometry-only beats our own hybrid on both metrics at $2.4\times$ the speed, which is the same null result for the vision detector that §5.4 reports on OHR-Bench.

What this measures is structural fidelity and not retrieval quality. A publisher wrote those sections to organise a document for a human reader, not to be retrieved, and a chunk that deliberately spans two sections may serve a question that needs both. Agreement with JATS is therefore evidence that a chunker recovers the document’s own structure, which is what we claim, and not evidence that structural boundaries are the right boundaries for RAG, which we do not claim and this experiment cannot show.

The ordering is stable under a $5.6\times$ increase in corpus size: a 24-document pilot gave the identical ordering with every value within 0.02, which matters because segmentation metrics on small corpora often fail to replicate.

We report both metrics because either one alone is quotable in the opposite direction. WindowDiff counts boundaries per window, so it punishes a tool that puts three splits where the document has one. Pk asks only whether a window’s two ends share a segment, so it forgives over-segmentation. The gap between the two grows with chunk count: LangChain at 86 chunks and unstructured at 107, against a gold averaging about 25 sections, are last under WindowDiff and mid-table under Pk.

Straddle rate is descriptive and not an error term. 57.4% of gold sections are shorter than MIN_CHUNK_TOKENS, so merging them is the size floor of §3 working as specified. The rate bounds how meaningful a per-chunk section *label* can be,

and our 0.513 is not a win to claim.

One qualification bounds the whole table: the ceiling is 84.7% of body sections locatable in the PDF’s own text, so every score reads against that and not against 100%. Unstructured’s 35 hard failures matched the pilot’s 6 of 24, so the rate is systematic and is itself a finding about that tool on born-digital PDFs.

5.3 Determinism, measured

Determinism is easy to assert and rarely tested past a single file, so we tested it. We parsed all 95 OHR-Bench documents twice, each parse in a separate process, and compared a hash over the full chunk structure: identifier, type, page, reading order, section path, and a content digest. Two runs agree only if every chunk matches on all six. Separate processes matter, because a same-process check cannot observe what varies across a process boundary, which is where non-determinism comes from: hash seeds, iteration order over addresses, thread scheduling, kernel selection.

All 95 documents were byte-identical across 5,810 chunks per run, with zero differing documents. That total is an unplanned cross-check: 5,810 is exactly the count recorded for DOCSTRUCT-GEO in the OHR-Bench run of Table 2, produced days earlier on different hardware through a different code path.

Two limits belong with the claim. Determinism holds within a version, so a persisted index must pin one. And this run is geometry-only: the hybrid path runs a model through CUDA, whose kernel selection is not guaranteed bit-reproducible, and we had no GPU available to test it, so the guarantee is established for DOCSTRUCT-GEO and remains unverified for DOCSTRUCT.

5.4 Where DOCSTRUCT loses

The vision detector does not generalise. It fails to pay for itself on any corpus we can check. On OHR-Bench the hybrid-versus-geometry ablation gives $+0.0012$ under span ($p = 0.80$) and $+0.0090$ under region ($p = 0.12$), neither significant on 3,558 questions. On the PMC section metric, geometry-only beats the hybrid at $2.4\times$ the speed (Table 4). We report it as a component that earns its cost on table-dense pages, and not as a contribution. Its one sign of life is the strict half of the threshold sweep ($+0.0092$ at 0.7): small, insignificant, and in the direction a table-heavy corpus would predict.

We are expensive in context. DOCSTRUCT retrieves 2,194 context words per query against unstructured’s 560, a $3.9\times$ larger context bill. Normalising rank quality by that bill, unstructured returns 1.166 MRR per 1,000 words against our 0.322, so it is $3.6\times$ more efficient per token even though it ranks 5th under span. We buy accuracy with tokens, and for a cost-sensitive deployment that trade may not be worth making.

We discard document text. On OHR-Bench our word coverage is 0.9632 against LangChain’s 1.000. Raw-text splitting drops nothing by construction, and we drop references, headers and footers by design. Our chunk duplication is $1.83\times$, the highest of the seven, coming from inline headers and separately emitted table chunks. DOCSTRUCT optimises retrieval and not raw preservation.

FinanceBench is not a retrieval benchmark for us. We intended FinanceBench (Islam et al., 2023) as a second external leaderboard, and it cannot serve as one under our protocol. Holding chunks fixed and varying only the embedder, the rank of the first relevant chunk was 81–299, 63–181 and 42–252 across three small encoders; Recall@5 was 0/5 and MRR@5 was 0.0000 under all three. The scoring rule is not at fault, since a relevant chunk exists at region overlap 0.99–1.00 and retrieval never reaches it. FinanceBench questions are analyst prompts with almost no lexical overlap with their evidence, that evidence is a numeric table which small encoders embed poorly, and a 10-K yields 729–1,176 chunks. A table of zeros reads as a chunking result when it is a retrieval limit. We report it rather than omit the corpus, because El Bachyr et al. (2026) reach MRR 0.700–0.844 on these same PDFs where our protocol reaches 0.0, which makes the usual caveat about cross-paper numbers concrete.

6 Conclusion

The result we would most like carried forward is that the leaderboard depends on the relevance rule, measurably and reversibly, on chunks that never changed. Since no rule is size-neutral, a comparison reporting one rule entangles a ranking with a size preference, and that entanglement can reverse the conclusion. Until reporting more than one rule is standard practice, chunker comparisons, ours included, should be read with the rule in view.

Two smaller results follow. Publisher-authored

JATS structure scores chunk boundaries with no retriever and no relevance rule in the loop, which sidesteps the question of who wrote the gold. And determinism, in a pipeline with no model on its parse path, is checkable in an afternoon. DOCSTRUCT itself wins where the rules favour large structural units and loses where they favour small ones: a narrower claim than “best chunker”, and the one the evidence supports.

Limitations

Scope. DOCSTRUCT reads PDF primitives, so scans without a text layer are out of scope and we run no OCR. IEEE two-column templates are unrepresented in our corpora, and every corpus is English.

Speed is our sharpest practical limitation. A figure-dense 18-page paper takes roughly 8 minutes to parse, dominated by materialising 422k vector primitives, and this is on the user-facing parse() path rather than only in the benchmark. The determinism run (§5.3) put a number on the tail: three of 95 documents exceeded a 30-minute per-parse cap and needed four hours to finish, all dense financial filings of 120 to 217 pages against a corpus median of 18. Page count alone does not explain it, since the corpus’s longest document at 382 pages parsed without trouble; table density does. Those three are the same filings FinanceBench is built from, so the cost is measured on two independent corpora, and it is a real barrier to using DOCSTRUCT on long financial documents today.

Two things we assert but do not measure. We report no edit distance or TEDS on OmniDocBench or READoc, so “the parse is correct” currently rests on retrieval quality instead of a direct measurement. And our layout metric rests on two hand-annotated documents: the DocLayNet validation split is its correct home and has not been run, which leaves several fusion constants marked # unvalidated in §3. Token-level IoU and Precision_Q (Smith and Troynikov, 2024) would replace the context-word column of Table 2, but our dumped runs record the overlap score behind each retrieved chunk without the chunk and gold token sets, so that metric needs a re-run.

Domain coverage. Table 3 shows we trail on academic and finance documents, and the academic slice is 10 documents, too thin to settle the question either way.

Ethics Statement

All three corpora are public: OHR-Bench, open-access PubMed Central articles under licences permitting reuse, and FinanceBench, built from public regulatory filings. None involves human subjects or personal data beyond author lists in published papers, and we redistribute no corpus. The pipeline invokes no language model, which bounds its energy cost to local CPU parsing. We report results in which our own system loses (§5.4) and a corpus on which our protocol fails entirely (§5.4), because selective reporting of chunking benchmarks is the specific harm this paper is about.

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